

KARLA VASQUEZ

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SKILLS

- **Languages:** TypeScript, JavaScript, Python, C/C++, SQL, Swift, HTML/CSS
- **Frameworks & Tools:** Angular, Tailwind CSS, SwiftUI, Astro, Firebase, Git, Unix, MongoDB
- **Spoken Languages:** English (Native), Spanish (Native)

EDUCATION

University of Southern California

Aug 2021 - May 2025

B.S. Computer Engineering and Computer Science

Los Angeles, CA

- **Relevant Coursework:** Data Structures & Object Oriented Design, Embedded Systems, Algorithms, iOS App Development, Full Stack Web Development, IoT, Computer Networks
- **Involvement:** Autonomous Networks Research Group, Association for Computing Machinery, Creative Labs, Women in Engineering, Viterbi Impact Program

WORK EXPERIENCE

Software Developer Intern

Sept 2025 - Present

Platia

Remote

- Migrated a legacy **Angular 6** to **Angular 19** application, improving maintainability through modular **TypeScript** architecture
- Rebuilt **responsive UI components** using **Tailwind CSS** and modern Angular patterns, enhancing usability across devices
- Integrated **RESTful APIs** for authentication, user management, and transactional workflows using standardized **HTTP services and error handling**

Academic Tutor (ELA & Math)

Sept 2025 - Present

HeyTutor

Soledad, CA

- Tutor **3-5 students** at a time in Math and ELA, adjusting pacing and explanations to individual learning needs
- Administer beginning, mid-semester, and final assessments to track progress and identify learning gaps
- Support a positive learning environment by motivating students and reinforcing productive behaviors

Undergraduate Researcher

May 2023 - Aug 2023

USC ANRG Summer Research Academy

Los Angeles, CA

- Conducted research on AI-driven personalized learning using **GPT-3**; authored a **sole-author abstract accepted to IForE 2023 Research Forum**
- Analyzed qualitative user interaction data weekly to identify engagement patterns, collaborating with faculty advisor to iterate research methodology

PROJECTS

MindSpark (Swift, SwiftUI, Anthropic Claude API)

April 2026

- Developed an AI-powered iOS learning app with **two practice modes (Math Blitz and Story Quest)**, integrating **Claude API** to generate adaptive hints and reading passages for elementary students
- Implemented **protocol-based AI service architecture** with mock and real implementations, enabling testable UI development
- Built persistent progress tracking with **UserDefaults** and **Codable** for stars, streaks, and multiplication mastery records, covering core functionality with **XCTest** unit and UI test cases

Personal Portfolio Website (Astro, TypeScript, SCSS)

March 2026

- Architected modular **Astro** components for portfolio, blog, and project case studies with **TypeScript** type safety and **SCSS** theming system

WanderPost (Swift, SwiftUI, Firebase)

May 2025

- Built a full-featured iOS postcard journal in **Swift/SwiftUI** using **MVVM architecture** with **MapKit** annotations and **Firebase** cloud sync

The SCAvenger (C, Embedded Systems)

Feb 2025 - April 2025

- Designed an environment-aware puzzle game with GPS-triggered challenges, achieving **60% RAM reduction** on a **2KB SRAM microcontroller** while integrating **4 I2C sensors** within a **\$100 budget**